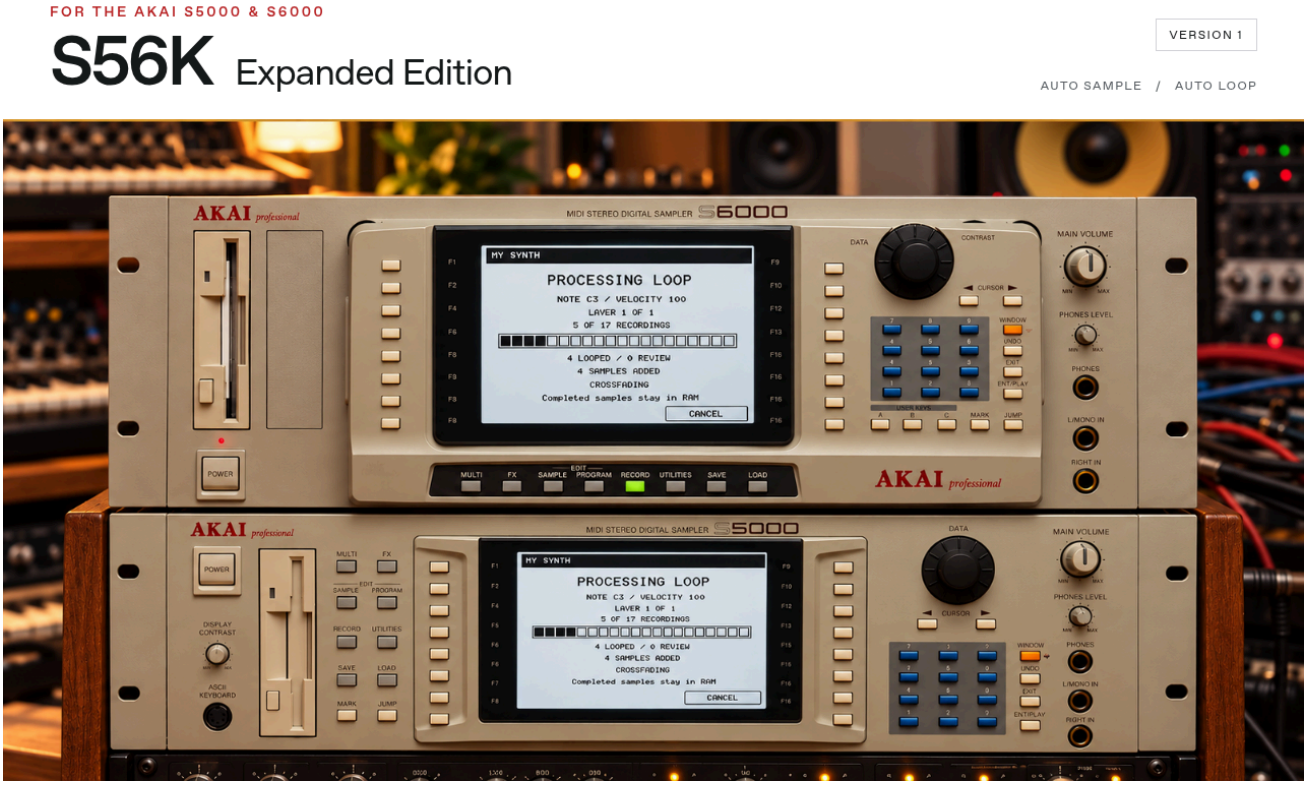
The user guide.

Auto Sample & Auto Loop
For the AKAI S5000 and S6000



Create. Capture. Play.
Turn an external instrument into a playable Akai program. Choose your notes and velocities, prepare the sustain, and save when you are ready.

Read the installation guidance, compatibility evidence and flashing disclaimer in the appendices before loading firmware.

Version 1.2 · 4 November 2026 · Independent project for AKAI hardware

Inside the guide

Create and name an empty program. Capture your instrument into RAM. Listen, refine and save the program with its samples through the normal SAVE page.

01 Connect and prepare

02 Set the capture range

03 Check levels

04 Capture velocity layers

05 Recording and Auto End

06 Loop the sustained part

07 Run, listen and save

08 Troubleshooting

09 Expanded Edition startup

Installation and reference

A · Installation B · USB driver & updater

C · USB reference & troubleshooting

D · Compatibility E · Flashing disclaimer

About this edition

S56K Expanded Edition adds automatic sampling and optional loop preparation to the AKAI S5000 and S6000. The visible EXPANDED EDITION V1.2 label identifies these additions; the underlying OS identity remains 2.14.

The screen illustrations use native firmware renders with example settings and progress states. The studio artwork is a presentation image, not evidence of hardware testing.

Firmware availability and exact-file instructions are published separately in the release area. Install V1.2 over USB with Firmware Updater 1.1 RC1, or from floppy. The new updater also retains Expanded V1.1, V1 and original Akai v2.14 support. This manual does not contain an OS image.

YOUR PROGRAM. YOUR WORKFLOW.

01

Create & name.

Start with an empty program.
Choose the sound to capture.

02

Capture into RAM.

Record, map and optionally loop.
Listen across the finished program.

03

Save when ready.

Use the normal SAVE page.
Your samples stay under your control.

1. Connect and prepare

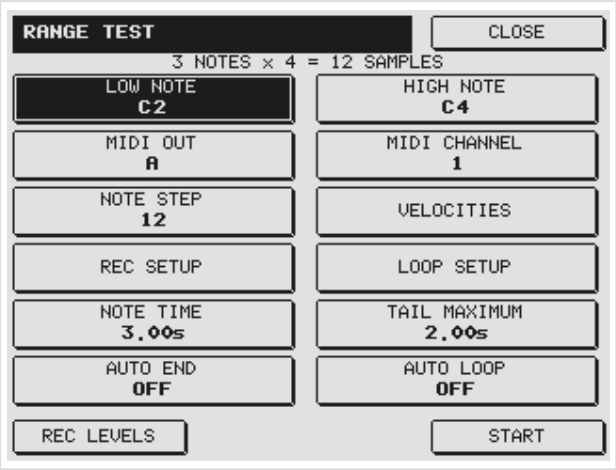
Connect the sampler's **MIDI OUT A or B** to the instrument's MIDI input. Connect the instrument's audio output to the sampler's recording input: left input for a mono capture, or both left and right for stereo. Select the appropriate input source using the normal recording controls.

Choose the sound on the instrument, set its receiving MIDI channel and leave that sound selected throughout the run.

Create an **empty program**, give it a useful name and open **Program Tools → Auto Sample**. The initial empty keygroup must contain no sample assignments. A run adds its own keygroups and velocity zones. To make a second run after completing or cancelling the first, start with another empty program.

Leave generous free sample RAM. Stereo recordings and multiple velocity layers use more memory, and loop processing needs temporary copies while preserving the original recording. During a run, close ak.Sys, the Editor and other sampler-control applications so they cannot request recording or memory operations at the same time.

2. Set the capture range



Press the soft key beside a field to select it, then use the **DATA wheel** or **+ / –** to change the value. The submenu buttons open their own pages; **BACK** returns to the main setup.

Main-page control	What it does
LOW NOTE / HIGH NOTE	First and last pitches to capture. Set both to the same note for a one-note test.
MIDI OUT	Selects physical output **A** or **B**, matching the rear-panel labels.
MIDI CHANNEL	Must match the instrument's receiving channel.
NOTE STEP	Distance between captured notes, in semitones. Smaller steps produce more recordings and less pitch shifting during playback.
NOTE TIME	Requested time to hold each outgoing note before sending Note Off.
TAIL MAXIMUM	Maximum recording time after Note Off, allowing the instrument's release and effects to fade.
AUTO END	Chooses whether quiet audio can shorten the held note, the tail, or neither.
AUTO LOOP	Enables optional loop processing after each recording.

The summary shows pitches × velocity layers = recordings. The high endpoint is included even when the final interval is shorter than Note Step. The limit is **99 captured pitches**, with up to four layers per pitch, subject to available memory.

Note names follow the Akai convention: **MIDI note 60 is C3**. Another instrument may call the same pitch C4; match the MIDI note number.

For example, C2 to C4 with a step of 12 captures C2, C3 and C4. With four velocity layers, that produces twelve recordings. The sampler assigns playback ranges between neighbouring captured pitches across the selected low-to-high range.

3. Check levels and choose mono or stereo

Open **REC LEVELS** before starting a long run. This uses the sampler's existing input, gain and meter controls. Set **Record Mode** to MONO or STEREO here; the choice is retained when returning to Auto Sample.

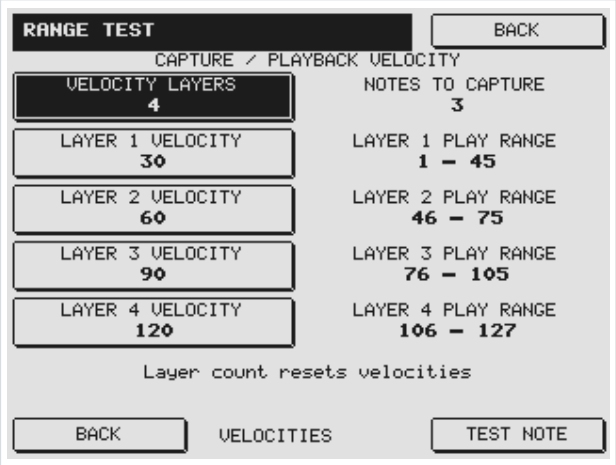
Use **TEST NOTE** and inspect the meters at the loudest velocity you plan to capture. Adjust the instrument's output and the sampler's recording gain for a healthy signal without clipping. For stereo, check both channels.

Auto Sample enables input monitoring while you are on its setup pages and during capture. It restores previous monitoring preferences when handing back control. Monitoring can pause during loop processing and between captures; another active recorder is not taken over.

Velocity may change the instrument's tone as well as its loudness. Recording gain controls the captured level. Set both appropriately for the sound.

START LEVEL, available in REC SETUP and through the native recording controls, determines when incoming audio triggers recording. Set it above the background noise but low enough to catch the intended attack. A slow or very quiet sound may need a lower threshold. Test one note and inspect its beginning before recording the entire range.

4. Capture velocity layers



Open **VELOCITIES** and choose one to four layers. Each layer specifies the MIDI velocity sent to the instrument. The displayed **PLAY RANGE** shows which incoming velocities will select that layer when you play the completed Akai program. Only the selected number of layers is displayed: with one layer, the other three rows are hidden.

Changing the layer count resets the capture velocities to a starting arrangement. Adjust them afterwards to suit the instrument. Velocities must increase from layer to layer; the controls maintain that order.

With four layers, the starting values are 30, 60, 90 and 120. Their playback ranges are divided at the midpoints, covering velocities 1–127 without gaps. Select a layer and use **TEST NOTE** to audition that velocity at the current low note.

Each pitch gets one keygroup, with its recordings placed in the appropriate velocity zones. Sample names include a shortened program-name prefix, the note and the capture velocity. Longer names are shortened to fit the sampler's naming limit.

5. Decide when recording should end



NOTE TIME covers the held note. **TAIL MAXIMUM** covers the sound after the external instrument receives Note Off. A sustained pad may keep sounding while held, then take several seconds to release. A guitar or piano sound may decay to silence while the note is still held.

AUTO END setting	Held note	Release tail
OFF	Uses the full Note Time.	Uses the full Tail Maximum.
TAIL ONLY	Uses the full Note Time.	Can stop early after confirmed quiet.
NOTE + TAIL	Can send Note Off early after the sound has been heard and then becomes quiet.	Can also stop early after confirmed quiet.

For sustained sounds that you intend to loop, **TAIL ONLY** is a useful starting point. It preserves the requested held duration while avoiding an unnecessarily long silent release. For naturally decaying sounds, **NOTE + TAIL** can save time across a large run.

When Auto End is enabled, REC SETUP exposes:

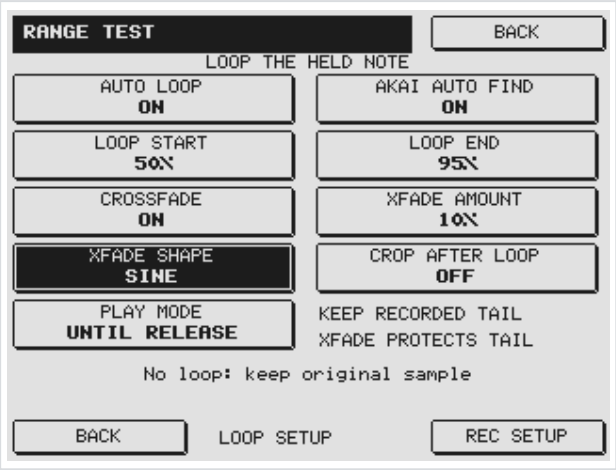
- **END LEVEL:** the level at or below which audio counts as quiet.
- **QUIET TIME:** how long that quiet condition must persist before it qualifies as an ending.
- **MINIMUM TAIL:** the earliest point after Note Off at which quiet can end the tail.
- **MINIMUM NOTE:** available in NOTE + TAIL mode; prevents quiet detection from shortening the held note before this time.

In stereo, **both channels must be quiet**. This helps retain a release or effect that continues on only one side. Longer Quiet Time is useful for sounds with rhythmic gaps, tremolo or delays.

The end detector uses the sampler's coarse meter scale. **METER FLOOR** is its lowest detection bucket, not a precision dBFS measurement. Choose a setting by listening to a test recording and observing the actual noise floor. A lower end threshold may preserve more decay, but noise can prevent it from being reached.

Unavailable controls disappear. Their values remain stored and reappear when the relevant mode is enabled. Auto End changes recording behaviour; it does not switch Auto Loop on or off.

6. Loop the sustained part



Enable **AUTO LOOP** and open **LOOP SETUP**. Its percentages refer to the **captured held-note portion**, starting at the sample's playback start and ending before the external release tail. They do not include the tail in their measurement.

Loop control	Meaning
LOOP START	Initial loop-start position, from 1% to 98% of the held portion.
LOOP END	Initial loop-end position, from 2% to 99%, above Loop Start.
AKAI AUTO FIND	Optionally refines the initial loop locally using Akai's existing routine.
CROSSFADE	Optionally processes the join using the native crossfade function.
XFADE AMOUNT	Requested crossfade length as a percentage of the resulting loop length.
XFADE SHAPE	Selects LINEAR, LOG or SINE.
CROP AFTER LOOP	OFF by default: keeps the full recorded release. ON trims after the loop end.
PLAY MODE	F6: UNTIL RELEASE loops while the playback key is held; IN RELEASE keeps looping after Note Off under the amplitude envelope.

For a steady synth tone, try a five-second Note Time, Loop Start 50%, Loop End 95%, Auto Find ON and Crossfade ON at 10%. These are starting settings, not a promise that every patch will loop cleanly. Aim for a stable section after the initial attack and before the release.

Auto Find OFF keeps the percentage positions as the starting loop without native refinement; the recording still has to pass the loop checks. **Auto Find ON** performs a local adjustment. It does not search every possible loop throughout the entire 50–95% region or automatically identify the musically best sustain.

Crossfade amount is limited by the available pre-loop audio and loop length. With **CROP AFTER LOOP OFF**, it is also limited to the held audio remaining after the loop end, so processing cannot alter the recorded release. The effective fade may therefore be shorter than requested. For a longer fade, try moving Loop End earlier. A larger percentage can smooth a join but may change the sound; audition the curves on separate test captures.

A positive crossfade can work with endpoints that do not already closely match. With Crossfade OFF, or when no positive fade will fit, stricter join checks apply. Amount and shape are hidden when Crossfade is OFF.

What happens to the tail?

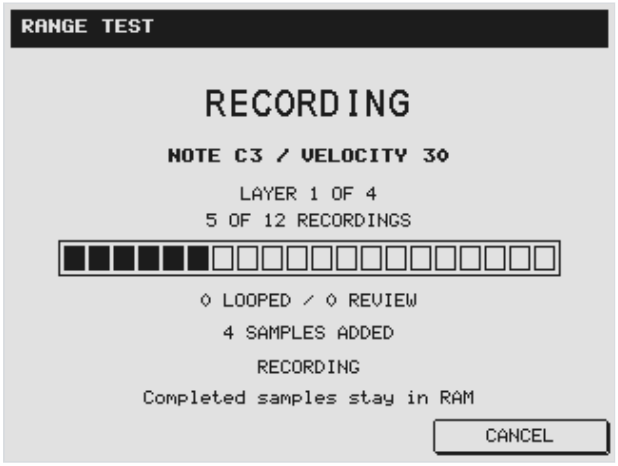
Result	Audio retained
AUTO LOOP OFF	The ordinary recorded note and tail.
Successful loop, CROP OFF	The attack, held note and full recorded release tail. Only pre-trigger material before playback start is removed.
Successful loop, CROP ON	The attack through loop end, plus the small native 15-frame playback margin.
Loop rejected for review	The original recording, including its recorded tail.

The sampler records the external release before processing the loop. Cropping does not skip that waiting period; AUTO END can shorten a quiet tail during recording.

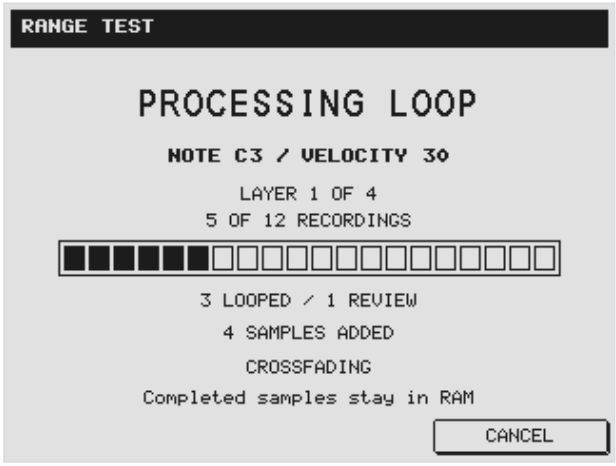
PLAY MODE defaults to UNTIL RELEASE. IN RELEASE allows looping to continue after the playback key is released. Auto Sample assigns program zones as **AS SAMPLE**, so they follow the mode stored in the sample. The amplitude envelope still determines what you hear after Note Off: retaining the recorded tail does not guarantee that every envelope setting will play all of it.

Auto Loop checks each channel and can leave a recording unlooped for review. Listen to these results in the ordinary sample editor. Evolving pads, heavy chorus, noisy textures and rhythmic material may require different positions or manual editing. Loop Setup contains settings for the recordings you are about to make. Use the ordinary sample editor to inspect their waveforms after capture.

7. Run, listen and save



Recording



Loop processing

Press **START** on the main page. The display follows preparation, waiting for audio, recording, release and sample assignment. When looping is enabled, it also shows the processing stage and counts of looped samples and recordings requiring review.

Between recordings, V1.2 can display WAITING FOR SAMPLE MEMORY or ORGANISING SAMPLE MEMORY. The sampler is waiting for background cleanup or arranging free recording space before the next note. This can take time, even when the resource page shows plenty of total free RAM. Let it finish; completed samples are retained. The sampler checks the available recording space again before continuing.

With **CROP AFTER LOOP OFF**, the final extraction stage is labelled **FINALIZING SAMPLE**, rather than CROPPING. It preserves the recorded release as described above; the label does not mean that cropping was enabled.

CANCEL stops the run while keeping completed sample assignments in RAM. Finishing an active recording or processing operation may take a moment, including when Cancel is pressed just after Start. If sample memory is being organised, cancellation waits for the owned cleanup to finish before another capture can begin. The current unfinished capture may not be assigned to the program; inspect the samples in RAM if the display asks you to check them.

If a loop edit is safely refused before processing begins, the original recording is retained for **REVIEW** and the batch can continue. An uncertain processing failure stops the run with a specific error. The display can show **LAST LOOP**, the affected processing stage, and an Akai error code when available. Photograph the whole screen for troubleshooting.

The capture summary can show memory wait, organisation and retry counters. A **LAST LOOP** review message can use that footer instead. If recording still cannot arm, the diagnostic screen preserves the failure details and counters; photograph it before leaving the page. A temporary native recording-memory error can be retried once after recorder cleanup. Other arm errors stop the run.

A restart-required fault keeps its explanation on screen and removes the inactive Cancel button. Further Cancel/Exit presses do not hide that message. Save completed work if the ordinary controls remain responsive before restarting.

After completion, play across the keygroups and velocity boundaries. For looped sounds, hold notes long enough to hear several repetitions, then check the release. Review any unlooped recordings and make ordinary sample or program edits as needed.

Save the program with its samples through the normal SAVE page. Auto Sample does not write your sound to disk. Unsaved recordings remain dependent on RAM and are lost when the sampler is powered off. Reload a saved program to confirm that the samples and loop settings were stored as intended.

8. Troubleshooting

What you notice	What to check
USE AN EMPTY PROGRAM	Create a new named program with no sample assignments. A completed or partial Auto Sample program is no longer empty.
MIDI plays, but there is no recorded audio	Check audio cables, input source, Record Mode, gain and Start Level. TEST NOTE and the meters help separate MIDI routing from audio routing.
NO AUDIO – CHECK START LEVEL	The audio did not reach the trigger threshold in time. Check the signal and threshold with a one-note test.
A recording is very quiet	Adjust audio output/recording gain while watching the meters. Check your chosen capture velocity too.
Decay is cut short	Lower End Level, increase Quiet Time or minimum timing, or use OFF to compare a full-duration recording.
The tail always uses its maximum time	Background noise may remain above End Level; in stereo, check both channels.
A loop is left for review	Try a steadier section, a longer held recording, different percentages or crossfade settings. The original recording is available for listening and manual work.
The saved tail is present but cannot be heard on playback	Check PLAY MODE and the program's amplitude-envelope release. With UNTIL RELEASE, looping stops when you release the playback key.
The crossfade seems shorter than requested	With Crop OFF, the fade is shortened to protect the recorded release. Try an earlier Loop End if more held audio is needed.
WAITING FOR SAMPLE MEMORY / ORGANISING SAMPLE MEMORY	Allow cleanup to finish. Total free RAM and immediately usable recording space are different. If you cancel, cleanup must still complete.
RECORD ARM FAILED	Photograph the diagnostic screen and record the settings, completed count and free-memory readings. Genuine insufficient RAM can still stop a run; save completed samples before clearing space or restarting.
FINALIZING SAMPLE, with Crop OFF	This is the release-preserving extraction stage. The recorded tail is kept; cropping after the loop has not been enabled.
INPUT MONITOR UNAVAILABLE	Check the recording source and whether another recording function owns the recorder. Monitoring can pause during loop processing and between captures.
An option has disappeared	Enable its parent option: AUTO END for quiet controls, AUTO LOOP for loop controls, and CROSSFADE for amount/shape.
A run stops with a processing or restart message	Stop further runs, inspect the samples already in RAM and record the settings/message for feedback. Save completed work through the normal controls if they remain responsive.

For useful feedback, include your sampler model, mono/stereo mode, source instrument, capture and loop settings, the displayed message and what you expected to hear. A photo of the relevant screen is helpful.

9. Expanded Edition startup



S5000 EX



S6000 EX

The boot screen shows a CPU/circuit graphic, the correct **S5000 EX** or **S6000 EX** model logo, and **EXPANDED EDITION V1.2**. The EX lettering is outlined to match the model design.

The separate **Operating System VERSION 2.14** field is retained for compatibility. Expanded Edition V1.2 identifies the added feature set; it does not change the underlying Akai version identity. Use the checksum accompanying a firmware download to identify its exact file.

Home · Quick start · Installation · Compatibility · Risk disclaimer

A · Installation

The firmware image is **S56K-EX.bin**, shared by the AKAI S5000 and S6000. The boot screen selects the correct model graphic automatically and displays **EXPANDED EDITION V1.2**. The underlying AKAI operating-system identity remains **2.14**.

Download V1.2 — S56K-EX.bin · SHA256SUMS.txt · What changed

Choose USB or floppy. Firmware Updater **1.1 RC1** recognises V1.2, V1.1, V1 and original Akai v2.14. USB installation requires an IB-S56USB card and Microsoft WinUSB. The older updater 1.0 RC1 still rejects V1.1 and V1.2; download the new app for those images.

Route	V1.2 support	What you need
Temporary floppy load	Runs V1.2 for that session without replacing the installed OS.	A readable floppy containing the verified BIN.
Floppy flash installation	Permanently installs V1.2.	A verified floppy and stable power.
Windows updater 1.1 RC1	Permanently installs the exact V1.2 BIN over USB.	Windows 11 x64, an IB-S56USB card, WinUSB and the current updater.

USB is not needed for Auto Sample. On **4 November 2026**, the project owner confirmed that flashing this exact V1.2 BIN worked, in response to the supplied flash-and-reboot test. The confirmation did not restate the sampler model or itemise each check; see the compatibility record.

Read the flashing disclaimer before proceeding. Permanent installation carries a risk of an unusable sampler and possible repair costs.

Before you begin

- Download the named firmware BIN and its checksum. Use **S56K-EX.bin**, not GitHub's automatic “Source code” archive of the entire repository.
- Back up your programs and samples to separate storage, and note customised settings. Flash installation can replace stored settings.
- Keep a known-working OS file and readable recovery disk. These do not guarantee recovery after a failed flash.
- Verify the BIN's SHA-256 checksum. The **1,048,576-byte V1.2** image is:

`e29e16a1c9dde87dd5ac543bed3ff2d195b56931765cc24b84ba7dfb4a767474`

In Windows PowerShell, use `Get-FileHash .\S56K-EX.bin -Algorithm SHA256`.

- **For floppy installation:** use a known-readable floppy, a reliable drive and stable power. Copy **S56K-EX.bin** to the disk with no other OS BIN files present. This is a file copy, not a raw disk image. Read the copied file back on the computer and verify its checksum again. Stop if it cannot be read or its checksum differs.
- Close ak.Sys, the Editor and other sampler USB tools. For floppy installation, disconnect computer USB; for USB installation, keep the updater's connection intact. The instrument's MIDI and audio connections can remain connected.

1. Load temporarily from floppy

1. Insert the prepared floppy and switch on the sampler.
2. Allow it to read the OS. Confirm **EXPANDED EDITION V1.2** appears on the boot screen. At **Write Flash OS**, select **NO** to continue without permanently installing it.
3. Check normal navigation. Open an empty test program, then **Program Tools → Auto Sample**. Make a short capture, audition it and check the controls you intend to use.
4. Save wanted work using the normal **SAVE** page before restarting.

If the V1.2 caption is absent or anything behaves unexpectedly, check the disk and exact file before considering installation. Choosing NO only dismisses the installation prompt; it does not load the BIN. The V1.2 caption confirms that the new image has already booted. A temporary boot leaves the existing installed OS in flash and does not prove that a flash write will succeed.

2. Install permanently from floppy

Proceed when the temporary load and checks have succeeded and you have accepted the installation risk.

1. Boot again using the same verified floppy and firmware file.
2. **On this boot, confirm EXPANDED EDITION V1.2 appears before selecting YES at Write Flash OS.** If the caption is absent or shows an older version, choose NO and stop. A flash prompt alone is insufficient: an older installed OS can also discover a BIN after a fallback boot. A successful temporary boot on the previous attempt does not confirm this boot's identity.
3. Allow writing and verification to finish. **Do not switch off, reset or remove media during the update.** Follow the sampler's completion instructions.
4. After successful completion, remove the floppy, switch off and restart. Confirm **EXPANDED EDITION V1.2** appears without the disk. This checks the OS stored in flash.
5. Check ordinary navigation and a short Auto Sample capture. Audition the attack, loop and release; save the program with its samples and reload it to check playback.

An update that reports an error is not a successful installation. Record the exact message before further action; do not assume a repeated update or a retained backup will recover every failure.

The normal Utilities **SAVE SETTINGS / SAVE TO FLASH** function stores settings. It does not install the OS. Use the boot-time **Write Flash OS** prompt above.

3. Install permanently over USB

Use **Firmware Updater 1.1 RC1** on Windows 11 x64. Start with a freshly booted sampler and no loaded libraries. Extract the app ZIP, connect the sampler through its IB-S56USB card and assign Microsoft WinUSB if necessary. The app's **USB driver setup** assistant can help identify the device and open Windows' driver wizard.

Select the sampler and click **Connect**, then choose the V1.2 BIN. Confirm that the app identifies **Expanded Edition V1.2** and reports an exact SHA-256 match. Click **Install firmware** and confirm. Keep power and USB connected until completion; there is no safe cancellation after erasure starts. Save the report, remove any firmware floppy and restart to verify the installed V1.2 caption.

The new app retains support for **Expanded V1.1**, **V1** and **original Akai v2.14**. The earlier physical USB round-trip used updater 1.0 RC1 with V1 and original v2.14 on an S6000 / Windows 11 Pro 25H2 setup. V1.2's hardware flashing confirmation used the floppy procedure; the new updater's software checks do not constitute a new physical USB test.

Supported USB files, driver setup and updater instructions →

The official **2.14** version number remains the same across these images. Verify V1.2 by its full **EXPANDED EDITION V1.2** boot caption after restarting **without an OS floppy**.

B · USB driver & updater

Firmware Updater 1.1 RC1 supports Expanded Edition V1.2, V1.1, V1 and original Akai OS 2.14. It checks the complete BIN against the approved checksum for that edition. If you still have updater 1.0 RC1, download the new app to install V1.1 or V1.2.

Windows 11 x64 · S5000/S6000 with an IB-S56USB card · Microsoft WinUSB. Extract the entire updater ZIP, then open `S56K-Firmware-Updater.exe`. Firmware is a separate download. This is an unsigned release candidate; see compatibility for test coverage.

Driver setup · once only

Already connects? Skip ahead. Otherwise, open **USB driver setup** in the app. The assistant lists present Akai devices and their current driver association, including devices that cannot yet be opened through WinUSB.

Select your sampler, then choose **Change Windows driver...** to open Windows' driver wizard. This explicit action requests administrator approval; ordinary updater use does not require elevation. In the wizard, choose:

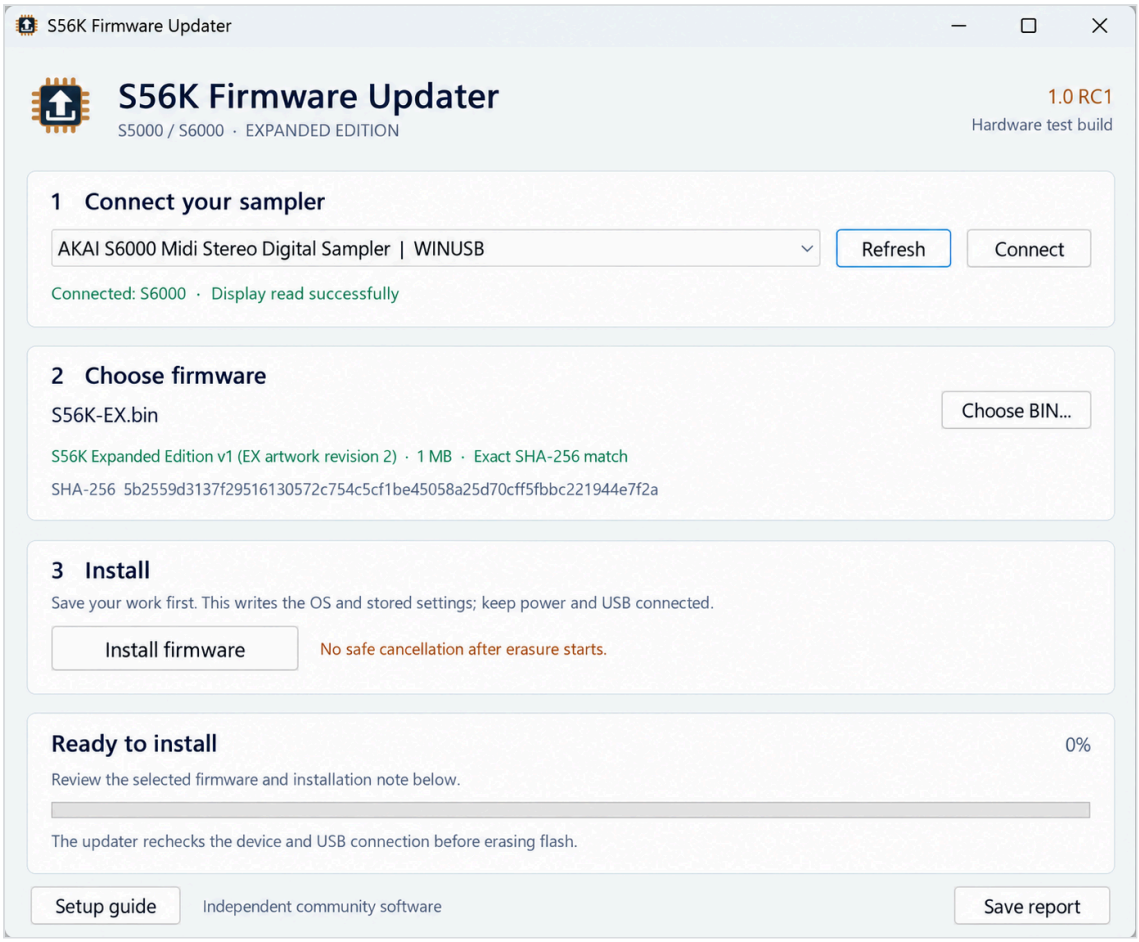
Browse my computer → Let me pick → Universal Serial Bus devices → WinUsb Device (Microsoft) → Next.

Check that you selected the Akai device (`USB\VID_09E8&PID_005E`). The assistant displays the local `winusb.inf` path when available for **Have Disk**. It does not silently choose a driver or install one on another device. Recheck afterward, close the assistant and click **Refresh → Connect**. The same steps remain available through Device Manager.

If WinUsb Device is unavailable or no usable interface is found, see the optional driver fallback. Changing the driver association can affect original ak.Sys access.

1. Connect and choose your BIN

Select the sampler and click **Connect**, then **Choose BIN...** and select the V1.2 firmware or another exact supported image. Look for **Connected**, the correct edition name, **Exact SHA-256 match** and **Ready to install**. V1.2 must be identified as **Expanded Edition V1.2**.

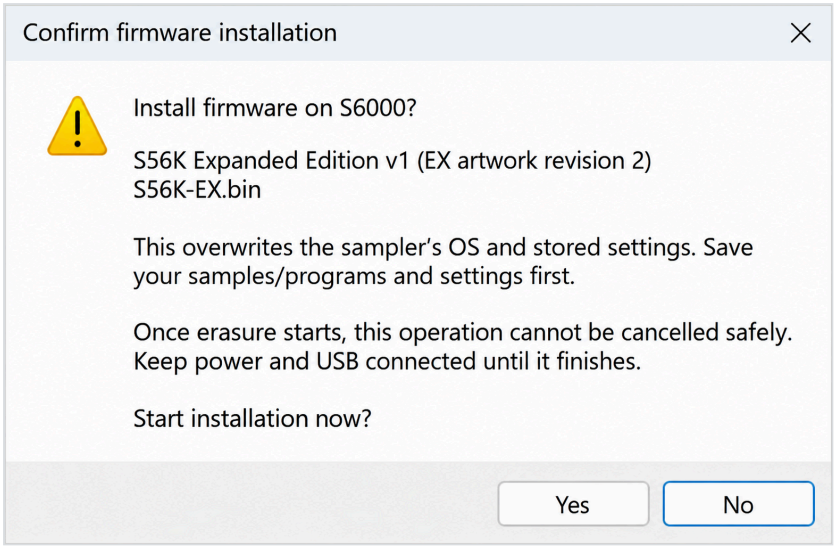


Illustrations show the same installation workflow using the earlier Expanded V1 app/screens. The current app identifies the selected edition; these illustrations are not evidence of a new hardware test.

2. Confirm the installation

Save your programs, samples and settings first. Start with a freshly booted sampler and no loaded libraries. Close the Editor, ak.Sys and other USB tools. Read the flashing disclaimer.

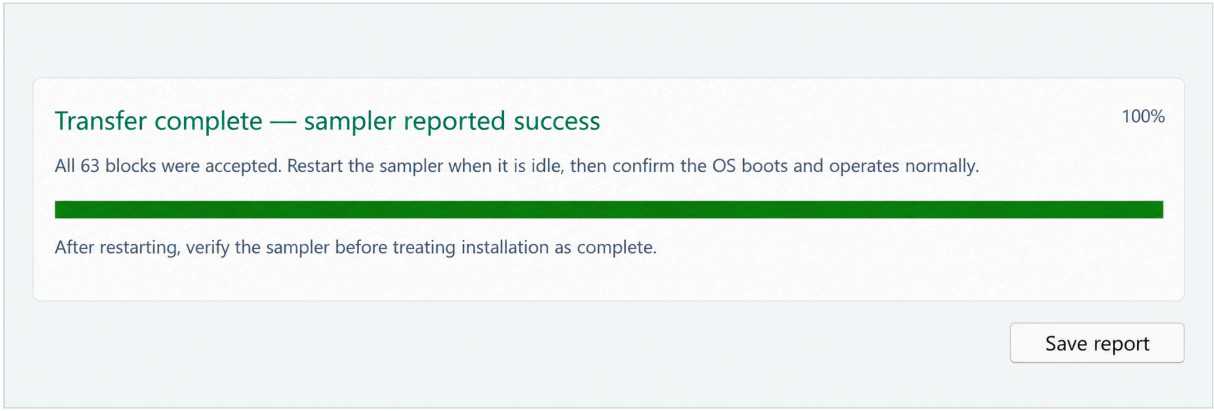
Click **Install firmware**, check the sampler and firmware shown, then choose **Yes** when ready.



This writes flash and stored settings. Keep power and USB connected until it finishes. There is no safe cancellation after erasure starts.

3. Finish, restart and check

Wait for **Transfer complete — sampler reported success.** Save the report.



When the sampler is idle, **remove any firmware floppy and restart it**. For the current firmware, confirm **EXPANDED EDITION V1.2** on startup and **Program Tools → Auto Sample**. If installing an earlier Expanded edition, check that edition's caption. For factory restoration, check the original Akai startup and ordinary operation. The underlying version field reads **2.14** for all supported images, so it does not identify the Expanded edition.

Close and reopen the app before another update. If an update reports an error after erasure starts, leave the sampler powered on initially and follow the failure guidance.

Checksums & supported files · Troubleshooting · Compatibility and test status

C · USB reference & troubleshooting

About this version

Firmware Updater 1.1 RC1 recognises Expanded Edition **V1.2, V1.1, V1** and **original Akai OS 2.14**. It also includes a USB driver setup assistant. The app remains standalone; the Editor does not perform firmware installation.

The app is unsigned. Windows may show an unknown-publisher or SmartScreen warning; check the download source and checksum rather than disabling Windows security or driver-signature enforcement. Windows 11 on Intel/AMD x64 is the supported host platform. Windows on ARM and other Windows versions have not been verified.

The earlier reported physical USB round-trip used updater 1.0 RC1, Expanded V1, an S6000 and Windows 11 Pro 25H2. V1.2's reported physical flash test used the floppy procedure. The new updater's software checks are described in compatibility; no new physical USB round-trip is claimed.

Driver fallback: WinUSB unavailable or no registered interface

Start with **USB driver setup** in the updater, or the Device Manager steps in the visual guide. Some Windows configurations need an additional interface registration. Use this fallback only when those steps are unavailable or leave the app reporting no registered interface.

1. Download **Zadig** from **zadig.akeo.ie** and run it. Driver installation normally requires administrator approval.
2. Choose **Options → List All Devices**, select the Akai sampler and confirm **USB ID `09E8 005E`**. Do not select a keyboard, mouse, hub or another device.
3. Choose **WinUSB** as the target driver and click **Install Driver** or **Replace Driver**. Do not choose libusb-win32 or libusbK for this app.
4. Close Zadig, reconnect USB and try **Refresh → Connect** again in the updater.

Both methods assign Microsoft's WinUSB to the sampler; Zadig is an optional setup helper, not a new driver requirement. See Microsoft's WinUSB installation documentation for interface-registration details.

What this changes: the Windows driver association, not the sampler's firmware. The updater's assistant opens Microsoft's driver wizard only after you explicitly choose the device and action; it does not silently replace drivers. You do not need to disable driver-signature enforcement. Assigning WinUSB does not make unmodified ak.Sys work; ak.Sys requires its separate compatibility setup and may stop recognising a device previously using its legacy driver. This updater needs neither ak.Sys nor its shim.

Supported files

All four supported firmware images are **1,048,576 bytes**. Renaming a recognised file is fine; editing it or using a different revision causes rejection. The app shows the edition identified by its complete SHA-256 checksum.

Firmware	SHA-256
Expanded Edition V1.2 — **`S56K-EX.bin`**	`e29e16a1c9dde87dd5ac543bed3ff2d195b56931765cc24b84ba7dfb4a767474`
Expanded Edition V1.1 — **`S56K-EX.bin`**	`cc3f6c056e0fb032303ee61f2e16c419c22708377cf888f0c1737be74321df93`
Expanded Edition V1 / EX artwork revision 2 — **`S56K-EX.bin`**	`5b2559d3137f29516130572c754c5cf1be45058a25d70cff5fbbc221944e7f2a`
Original Akai OS 2.14 — **`s56v214.bin`**	`05e466a7a19e5abde1e53660e0069bf2fb34bb33f705841185883a78b5369402`

The factory BIN is recognised for restoration but is not included in these downloads. Retain your original Akai image if you need that option. Unknown, corrupted and private test images remain rejected. The old 1.0 RC1 app accepts only the last two rows; use **1.1 RC1** for V1.1 or V1.2.

To check a download in PowerShell, run `Get-FileHash .\S56K-EX.bin -Algorithm SHA256` and compare the result with the table or the release's `SHA256SUMS.txt`. The **updater 1.1 RC1 ZIP** SHA-256 is:

`912ee879386676ac6a159a271e7d1f27115988a51bf644eb254224be806372c1`

Updating from factory OS: the running sampler OS performs the flash write. Factory v2.14 retains its older allocation and final-byte comparison limitations until the new OS is running. Adding new target images does not change the OS that performs the installation. Always start with a freshly booted, unloaded sampler.

Troubleshooting

What you see	What to do
No sampler listed	Check power, the IB-S56USB card, USB cable and direct PC port. Open **USB driver setup** to find present Akai devices and inspect the driver.
Driver missing / WinUSB not assigned	Select the Akai device in the assistant and choose **Change Windows driver...**. Follow the visual guide for `09E8:005E` only.
No usable interface registered	Use the WinUSB fallback above, reconnect USB and refresh.
Cannot open / access denied / connection busy	Close the Editor, ak.Sys and other sampler USB tools. Reconnect USB and try Connect again before installation.
Timeout during Connect	Save the report. Check cable and power; close competing tools. If needed, save sampler work and restart it before trying again. This advice applies before an update, not during flash writing.
V1.1 / V1.2 BIN rejected	Check that the app says **1.1 RC1** and compare the BIN's full checksum above. Replace an older 1.0 RC1 app with the current download; changing the Windows driver will not fix a file-check failure.
Other BIN rejected	Check it against the four supported hashes above. Do not select a private test build, ZIP file or GitHub “Source code” archive.
Install disabled after a completed update	This is intentional. Close and reopen the app for a new session.

If an update fails after erasure may have started: leave the sampler powered on initially, save the report and note its display. The OS in flash may be incomplete. Do not repeatedly retry or assume a power-cycle will fix it. Share the error with the project owner before further attempts; recovery may require the established floppy procedure. The updater cannot repair a sampler whose application cannot start USB services, and preserving the boot block does not guarantee recovery.

Use **Save report** to export the session. The app also attempts to save logs under `%LOCALAPPDATA%\S56K\FirmwareUpdater\Logs`. Reports contain device identifiers, file paths, firmware hashes and status; review them for personal paths before posting publicly. Nothing is uploaded automatically.

Download releases · Full compatibility notes · Report an issue

D · Compatibility

S56K Expanded Edition V1.2 is designed for the **AKAI S5000 and S6000**, using their shared v2.14 operating-system foundation. One model-aware firmware image supplies the features and appropriate startup artwork for each sampler.

Auto Sample uses the sampler's MIDI outputs and recording inputs. USB is not required for the sampling workflow. The visible **EXPANDED EDITION V1.2** label identifies this project's additions; the underlying OS/settings/protocol identity remains **2.14**.

Current downloads

Download	Supported use
Expanded Edition V1.2 — S56K-EX.bin	Shared S5000/S6000 firmware. Temporary floppy loading, permanent floppy installation or USB installation with updater 1.1 RC1.
Firmware Updater 1.1 RC1	Standalone Intel/AMD Windows x64 app; IB-S56USB card and WinUSB required. Accepts the exact published Expanded V1.2, V1.1 and V1 images, plus original Akai v2.14. Includes USB driver setup assistance.

V1.2 firmware SHA-256: `e29e16a1c9dde87dd5ac543bed3ff2d195b56931765cc24b84ba7dfb4a767474`.

Updater 1.1 RC1 ZIP SHA-256: `912ee879386676ac6a159a271e7d1f27115988a51bf644eb254224be806372c1`.

Use updater 1.1 RC1 for V1.2 USB installation, or use floppy. The older 1.0 RC1 app accepts only Expanded V1 and original Akai v2.14. Each supported BIN is checked against its full SHA-256; unknown or modified images remain rejected.

Verification evidence

Area	Evidence and limits
V1.2 floppy installation	On **4 November 2026**, the owner confirmed “flashing works” in response to the supplied V1.2 flash-and-reboot test. The confirmation did not restate the sampler model or itemise each check. It does not separately establish flashing on both models or installation over USB.
V1.2 software verification	All **20** software-verification suites passed for the exact published BIN, followed by independent review. They include memory recovery, cancellation, startup, both model boot-render paths, recording, native loop processing and flash paths. Bounded emulation and peripheral fixtures do not reproduce every hardware or timing condition.
V1.2 pre-release operation	The owner reported that the preceding private D2 recovery build was stable and Record Arm was working. V1.2 changes only two version-label strings from D2; instructions and memory layout are identical. No exact run counts or durations were supplied in that confirmation.
V1.1 floppy installation	On **2 November 2026**, the owner reported “Flashing worked” after receiving the exact V1.1 BIN and floppy-only test procedure. The report did not restate the sampler model or separately confirm each cold-boot/checklist step. This is a successful reported flash, not separate evidence for both models.
V1.1 software verification	All **16** software-verification suites passed for the exact published BIN, followed by independent review. Checks cover startup, layout, both model boot-render paths, recording, monitoring, looping, cancellation and installer behavior. Bounded emulation and peripheral fixtures do not reproduce every hardware or timing condition.
V1.1 pre-release operation	The owner reported the preceding private T2 build working very well. V1.1 changes only its boot caption and text pointer; executable code and memory layout are identical. The feedback did not restate the model or itemise individual tests.
Earlier Auto Sample on S5000 and S6000	Successful operation was reported on both models during feature development. This historical result does not separately establish V1.2 flashing on both.
Updater 1.0 RC1: earlier hardware test	On **23 October 2026**, the owner confirmed an **Expanded Edition v1 → original Akai v2.14 → Expanded Edition v1** USB round-trip after receiving the reboot and visual-check procedure. The established setup was **S6000 / Windows 11 Pro 25H2**. No separate S5000 USB, V1.1 USB or V1.2 USB result is established by that test.
Updater 1.1 RC1: software verification	All four approved firmware images passed exact-file validation; 36 altered-image cases were rejected. All 32 host/native-firmware cases passed, covering 16 running/target combinations and 16 transfer-failure cases, with startup checks for all four resulting images. The actual Windows executable passed file-selection, edition-label, rejection, report and driver-assistant checks under Wine. WinUSB wrapper and driver-helper fixtures also passed. These are controlled software checks, not physical USB or flash tests.
Updater 1.1 RC1: hardware scope	No new physical USB round-trip or real Windows driver reassignment is claimed for this app. The V1.2 hardware flashing confirmation above used the supplied floppy procedure; the earlier USB result belongs to updater 1.0 RC1.

Area	Evidence and limits
Windows / ak.Sys installation	The standalone updater does not require ak.Sys. Successful use of it does not verify firmware installation through ak.Sys.
Other hosts and configurations	Windows on ARM, other Windows versions and wider USB-card/host combinations have not been verified. The current updater is unsigned and remains a release candidate, version 1.1 RC1.

These hardware results are owner reports. They are not independent flash readback, recorded failure-recovery tests or guarantees for every machine. The published V1.2 BIN is byte-for-byte the file supplied for the reported flash test; its identity is established by the full SHA-256 checksum.

The running sampler OS performs erasure and programming. Factory v2.14 retains its older allocation and final-byte comparison limitations until the new OS is running. Start an installation with a freshly booted sampler and no loaded libraries. Flashing can replace stored settings, and there is no automatic rollback or guaranteed recovery. Read the flashing disclaimer.

Before a capture session

- Create a new, empty program and name it before opening Auto Sample.
- Leave generous free sample RAM. Longer recordings, stereo, additional velocities and temporary loop-processing copies need more space.
- Test the input level and attack on one note before capturing a large range.
- Close ak.Sys, the Editor and other sampler-control applications during Auto Sample to avoid concurrent recording or memory operations.
- Save the program **with its samples** through the normal SAVE page before switching off.

Auto Loop is optional. Results depend on the sound; recordings that fail normal loop checks retain their original audio for review. With **CROP AFTER LOOP OFF**, successful processing preserves the full recorded release. The program's amplitude envelope still determines the audible release. Crossfade may be shortened to protect that audio. See the loop guide.

V1.2 reserves **8 KiB more CPU RAM than V1.1**; sample RAM capacity is unchanged. It waits for background sample-memory cleanup and can request native memory organisation when recording workspace is insufficient. After cleanup and a fresh space check, a specific temporary recording-memory error may be retried once. Completed recordings are retained. This corrects the observed Record Arm failure path; genuine insufficient sample RAM and other recorder errors can still stop a run.

Memory organisation can take time, and cancellation waits for the owned cleanup to finish. If cleanup cannot be confirmed and a restart-required fault is displayed, do not start another capture. Save completed work through the ordinary controls if they remain responsive, and restart. Input monitoring can pause during processing and between recordings. The original allocator can also wait when its internal memory pools are exhausted; V1.2 does not replace that allocator or guarantee recovery from every memory condition.

Loading and installation

The installation guide separates temporary floppy loading from permanent floppy installation. Confirm **EXPANDED EDITION V1.2** on the actual installation boot before choosing **YES** at **Write Flash OS**. The prompt alone can also appear when an older installed OS has booted.

After successful permanent installation, remove the firmware floppy before restarting to verify the image stored in flash. The official **2.14** version field does not distinguish these editions: check the full V1.2 caption.

The USB guide covers driver setup and V1.2 installation using updater **1.1 RC1**, including its exact-file checks and reboot verification. The same app retains support for the earlier approved images.

About the images

The screen gallery uses native firmware renders at the original **320 × 240** resolution. Program names, settings and progress figures are examples; actual physical LCD appearance varies.

The landing-page studio image is presentation artwork. It illustrates the two sampler models and workflow, rather than documenting a hardware test or measured audio result.

E · Flashing disclaimer

Installing or using S56K Expanded Edition or the S56K Firmware Updater is your choice and is undertaken at your own risk. Read this notice and the instructions supplied with the specific firmware and updater releases before loading or installing them.

Understand the risk

S56K Expanded Edition is an independent modification for vintage AKAI S5000 and S6000 samplers. Writing firmware to the sampler's flash memory changes its installed operating system. An interrupted or unsuccessful update, incompatible hardware, failing media, power loss, or an undiscovered software fault may leave the sampler unable to start or operate correctly. Recovery may require specialist equipment, repair, replacement parts or service, and may not be possible.

Loading firmware temporarily and obtaining a successful result do not guarantee that a later flash operation will succeed. Testing, checksums and retained backup files reduce some risks; they do not make an update risk-free or provide automatic recovery.

Your responsibility

You are responsible for deciding whether to use the firmware and updater, checking the applicable model and release instructions, verifying downloaded files, backing up programs, samples and configuration, using suitable media, USB connections and a reliable power supply, and determining whether you have the knowledge and resources needed to proceed. Assign a USB driver only to the correctly identified sampler; changing a driver association may affect other software that uses that device. Do not interrupt an update once flash writing has started. If you are uncertain about the process or cannot accept the possibility of loss or repair costs, do not install it.

No warranty

To the maximum extent permitted by applicable law, the firmware, updater software, documentation and related materials are provided **“as is” and “as available,” without warranties of any kind**, whether express, implied or statutory. This includes, where applicable, warranties of merchantability, fitness for a particular purpose, satisfactory quality, accuracy, non-infringement and freedom from defects. No promise is made that the software will be error-free, meet your requirements, work with every hardware configuration, or allow recovery after a failed update.

Limitation of liability

To the maximum extent permitted by applicable law, the project owner will not be liable for any loss or damage arising from downloading, loading, installing, using or being unable to use the firmware, updater software or related materials. This includes damage to or loss of use of equipment; loss or corruption of programs, samples or other data; repair, replacement or recovery costs; loss of income or business; and direct, indirect, incidental, special or consequential loss, whether the claim arises in contract, tort or otherwise, even if the possibility of such loss was known.

Nothing in this notice excludes or limits liability that cannot lawfully be excluded or limited, or affects rights that cannot be waived under applicable law. Where a limitation is not permitted, it applies only to the extent the law allows.

Independent project

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